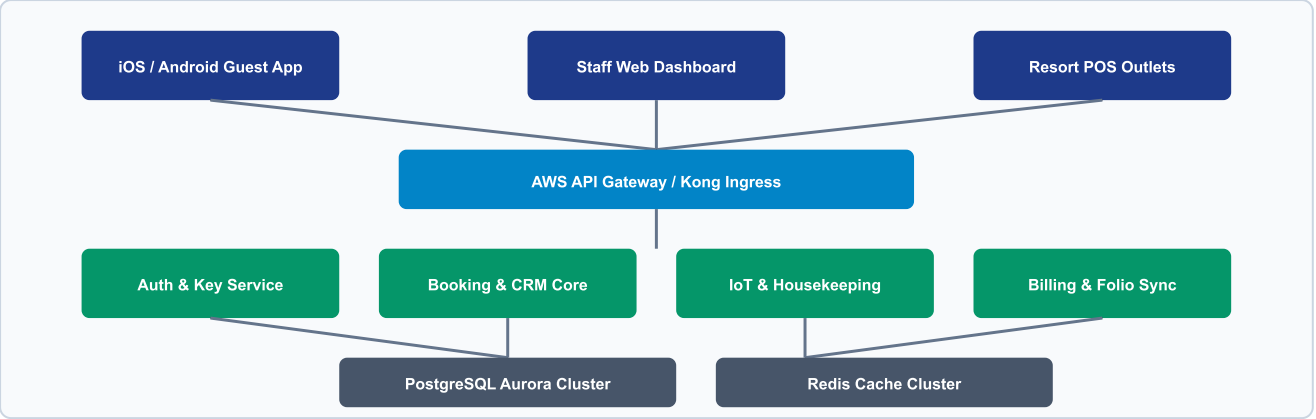
PROJECT DESIGN PHASE — ARTIFACT 10

High-Level Solution Architecture

Cloud Infrastructure, Microservices Topology & Enterprise Integrations

1. SYSTEM ARCHITECTURE BLUEPRINT



2. ARCHITECTURE HIGHLIGHTS & RELIABILITY DESIGN

- **Decoupled Microservices:** Independent deployment pipelines ensure high availability.
- **End-to-End Encryption:** Digital key generation utilizes AES-256 tokens sent directly over TLS to BLE door nodes.
- **Failover Resilience:** Dual-region database replication ensures zero downtime during regional AWS outages.